

Electrical



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electrical Engineering
Academic Year: 2025-26 (semester II)
In-Semester Examination II

Class: Final Year

Course Name: HVDC and FACTS

Course Code: PEC-EE410

Date: 24.03.2026

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	An HVDC converter operates at a firing angle of 20° and the reactive power consumed is 300 MVAR. If the firing angle increases to 30° , explain qualitatively how the reactive power demand changes.	10	CO3	BT2
Q.2	Explain the different types of harmonic filters used in HVDC transmission systems. Classify the filters into AC and DC side filters and describe in detail the construction, working principle, and application of each type such as single-tuned, double-tuned, and high-pass filters. Also discuss their advantages, limitations, and role in improving power quality in HVDC systems	10	CO4	BT4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electrical Engineering
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (Semester II)

Class: Final Year B.Tech.

Course: Electrical Machine Design

Course Code: PCC-EE403

Date: 25.03.2026

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q. N.	Attempt all questions.	Marks	CO	BT
Q. 1	Draw the neat diagrams of the laminations used in the stator and rotor of a small induction motor.	5	CO3	BT1
Q. 2	A 3 phase 4 pole induction motor has 24 slots. Calculate the order of slot harmonics produced. It is desired to eliminate the higher order slot harmonic. Find the angle through which the bars must be skewed.	5	CO3	BT3
Q. 3	Write down the expressions used to calculate the mmf for stator and rotor core of three-phase induction motor.	5	CO4	BT1
Q. 4	A 50 kW, 3000 V, 50 Hz, 8 pole, 3 phase star connected induction motor has magnetizing current which is 25 % of the full load current. Calculate the value of stator turns per phase if the mmf required for flux density at 30° from pole axis is 400 A. Assuming winding factor = 0.95, and full load efficiency = 0.90 and power factor = 0.85.	5	CO4	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Nanded**

**Department of Electrical Engineering
In-Semester Examination I
Academic Year: 2025-26**

Class-B -TECH

Subject: Power Quality and Harmonics

Subject Code – PCC-EE414

Date: 24/03/2026

Time:14:30 to15:15

Max. Marks: 20

Note

- i. **ALL QUESTIONS ARE COMPULSORY**
- ii. **THIS QUESTION PAPER CONTAIN ONLY 01 PAGE**

<i>ALL QUESTIONS ARE COMPULSORY</i>		Marks	CO	BT
Q.1	Explain common grounding problems in electrical installations and their effects on power quality	6	CO3	BT-2
Q.2	Explain the working principle and uses of a power quality analyzer in monitoring electrical disturbances	4	CO3	BT-2
Q.3	Explain the effects of harmonics on power system apparatus such as transformers, motors,	6	CO4	BT-2
Q.4	Explain methods used to minimize or control harmonics in electrical systems.	4	CO4	BT-2

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create